

Evidentiary reasoning within the Analytical Hierarchy Process: Use of graph entropy to rank decision elements

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Abstract

Decision makers often require insight into the relative strength of the elements of evidence on which their choices hinge. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of elements' relative strength. Comparative judgements ("pairwise comparisons") between elements of a given hierarchy model produce lists of edges for a complete graph. Under certain conditions, such graphs are also directed and acyclic, belonging to the class of tournament graphs. Intensities of Importance from comparative judgements provide weights for graphs' edges. Relative-entropy calculations based on Laplacians for induced subgraphs quantify the relative evidentiary strength associated with the decisional evidence omitted from them. A previously published case study related to an offshore-pipeline material selection demonstrates the approach. The graph-entropy method provides greater strength-of-contribution contrast than sensitivity analysis from the case study.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

"...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to a pretence of exact knowledge that is likely to be false..." (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge confronting multiple facets. We conventionally decompose uncertainty into *epistemic* and *aleatory* (or *stochastic*) components (Salicone & Prioli (2018)). Sound evidentiary reasoning must account for the consequences of both.

The present study focuses primarily on the *epistemic*. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making.

Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Townsend, et al (2025); Ehrig, et al (2026), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is our evidence?

Ehrig, et al (2026) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of very small samples of data" (emphasis added). Circumstances demanding this sort of reasoning arise in numerous contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to *epistemic* explorations (Zellweger & Zenger (2023); Camuffo, et al (2024a)). Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when

trying to select optimum scenarios.

Nontrivial challenges inevitably arise even when considerable amounts of statistically coherent, phenomenologically illuminating empirical data exist. Cover & Thomas (2006) derive an information-theoretic data-processing inequality: “...no processing of Y ... can increase the information that Y contains about X .” Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive non-stationarity in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2025) articulate the case for “...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems...” for many tasks.

Extending Dempster’s (1967) seminal concept, Shafer (1976) offered a rigorous framework to characterize epistemic uncertainty in evidentiary reasoning. More-recently, Shenoy (2023) applied Shafer’s reasoning approach to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024b); Ehrig, et al (2026)).

Hu  , et al (2026) used a stochastic-detection area-under-the-curve (AuC) approach (cf. Fawcett (2026)) to ascertain evidentiary elements’ relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for “...logic and reasoning with the aid of very small samples of data” (Ehrig, et al (2026)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible evidentiary span: The range of measurable factors — elements in the vernacular of Saaty (1986, 1994) — believed to differentiate within the range of scenarios under consideration. A *hierarchy model* represents such spans as tree graphs. The responsible individual

decision maker or a panel of experts then performs *comparative judgements*: “...the numerical representation of a relationship between two elements that share a common parent” in a tree graph (Saaty (1994)). A case study herein demonstrates AHP’s use to rank plausible alternatives.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025) and S  enz-Royo & Chiclana (2026) studied methods to improve the robustness of expert inputs into comparative judgements. Syed, Shabarchin, Lawryshyn (2020) describe AHP’s use in eliciting prior probabilities for Bayesian reliability analysis.

Evidentiary-strength analysis using the AHP.

That comparative judgements for an AHP hierarchy model define a complete graph provides the seminal point of departure for this investigation. Additional opportunities arise when the resulting graph is also directed and acyclic. Saaty (1986) recognized this: “Partially ordered sets with a finite number of elements can be conveniently represented by a directed graph”. Complete Directed Acyclic Graphs (DAGs) belong to a special class of graphs called *tournament graphs* (Moon (2015)). Nishizawa (1999) obliquely considered graphical methods for AHP consistency. The decision-theory literature however contains scant exploration of AHP’s graphical characteristics and opportunities arising therefrom.

Entropy — a key concept from information theory — provides an abstract but rigorous framework for characterizing evidentiary strength (Cover & Thomas (2006)). In general, “the entropy of a graph is interpreted as its *structural information content* and serves as a complexity measure” (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Structural information content bears direct relevance to the present concern of evidentiary reasoning. Brown, et al (2020) studied the particular characteristics of tournament graphs.

What questions do entropies of AHP-based graphs allow us to answer? Endogenous to the AHP formulation itself, relative entropy quantifies the marginal informational contribution by a subset of elements within the

evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures how- ever do not require extensive empirical data spanning of both positive and negative cases, as with Hué's ap- proach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of *belief-function graphical models* to decompose their elements' relative evidentiary strength.

Evidentiary strength relates to the concept of *Value of Information* on which the disciplines of Information Economics (Birchler & Büttler (2007)) and Decision Anal- ysis ((Howard & Abbas , 2016, Chapter 18); Abbas & Hazen (2025)) are substantially based. The present in- vestigation proposes an information-theoretic perspec- tive. In contrast, information economics and decision analysis focus on utility values of information. Notwith- standing, Birchler & Büttler (2007) do invoke additional concepts including information theory and probabilistic state spaces (along the lines of de Finetti (1974)).

By what mechanism do we infer the differential in- formation between hierarchically-adjacent AHP hi- erarchy models? Relative entropy provides a well- understood approach (Cover & Thomas (2006)). Rela- tive entropy for graphs has been well-researched (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xi- aotong (2025)). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost by omission of elements from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for complete, mixed graphs with weighted edges. Brown, et al (2020) analyzed directed graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graphs comprised of weighted edges. The intensities of importance from the AHP comparative judgements provide graph-edge weights used here.

Second, relative entropies of hierarchically adjacent hierarchy models must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into constituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a complete graph — for each hierarchy model. (Cover & Thomas , 2006, §2.5) describe *chain rules* for entropy, relative entropy, and mutual informa- tion. Such mechanisms offer a rigorous approach to link

hierarchically coupled graphs.

Why is “heavy” machinery of graph entropy called- for? One might intuitively assume that individual com- parative judgements suffice to inform us about the most- influential elements. The AHP's machinery however in- troduces “mutual coupling” between elements: “...many of the concepts derived for hierarchies also relate to sys- tems with feedback” (Saaty (1986)). That comparative judgements define a graph — a network — provides the basis of their power. Our graph-entropy machin- ery accounts for the aggregate effect of all the evidence informing a decision or risk assessment. Individual im- portance intensities fail to provide this.

Decision Quality as a check on the complete- ness of evidentiary reasoning.

Decision Quality (DQ) represents a prominent applied decision-analysis framework that emphasizes eviden- tiary reasoning (Spetzler, Winter, Meyer (2016)). Philo- sophically congruent with design-of-experiments prac- tice (Cox & Reid (2000)), DQ provides a *de facto* stan- dard of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. Introducing DQ here — independent of the mechanics of AHP — reinforces our logical com- pleteness.

The pattern of reasoning underlying Saaty's (1986) AHP methodology loosely maps to the DQ framework. Figure 1 depicts the framework, comprised of six links.

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with proscriptive judgements to provide analyti- cal clarity. Saaty (1994) described this as “...the most creative part of decision making that has a significant effect on the outcome...”.
- By definition, the act of deciding requires selection of one or more options from a discrete, precisely defined set of *Creative Alternatives*.
- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified. They specify the decision makers' values, clarifying why the best alternative actually is the best.
- *Sound Reasoning* defines the logic employed in identifying the best alternative given the available evidence. Creighton (1909) cautioned that “...it

Figure 1: Decision-Quality framework. (From Society of Decision Professionals (DQ))



is not possible to lay down definite rules for the guidance of thinking in every case.” The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach, with accompanying advantages and disadvantages.

- **Commitment to Action** represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to act, a decision arguably has not occurred. The decision analysis was merely academic. Persistence with the status quo nearly always occurs among the alternatives under consideration. Such a selection should however result from applying sound reasoning to the relevant available evidence.

Now, the use of AHP by no means suffices to achieve DQ. Thoughtful application of AHP does address a subset of DQ’s links. Injudiciously applied, it does not however guarantee satisfaction of the von Neumann-Morgenstern axioms (Kochenderfer, 2015, p. 58) undergirding decision science. Featuring AHP as an approach to DQ in no way constitutes our objective here. We instead explore AHP’s ability to highlight how subsets of decisional values and spans of evidence influence the results of sound reasoning.

Course of investigation.

We begin with a brief illustration of AHP methodology. A *real-world* case study concisely demonstrates its application to preference-ranking a set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The complete graphs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement graphs associated with distinct hierarchy models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes weighted directed edges and unit-weight undirected edges. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted directed-graph Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based graph structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entropy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

AHP case study .

Zeng, D., et al (2025) employed AHP to select materials from which to construct pipelines for use in offshore oil and gas extraction. Their analysis considers metallurgical technical details that are tangential to our purposes. We principally focus on the AHP mechanics.

Zeng, D., et al (2025) particularly suits the purposes of this investigation for three reasons.

1. *It demonstrates a real-world application.* Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration. It involves nontrivial, idiosyncratic complexities that challenge our methodology.
2. *This case is recent.* Given that the AHP literature spans decades, a reader might succumb to the temptation to conclude that the method is anachronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically meaningful application.
3. *It strikes a balance between tractability and complexity.* Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demonstrate the use of relative entropy to couple hierarchically adjacent hierarchy models. That it considers nine distinct criteria also makes it interesting.

The following offers neither criticism nor endorsement of Zeng, D., et al (2025). Their work merely serves to illustrate AHP mechanics.

We strive in the following to select AHP terminology that remains close to Saaty (1986, 1994). Variations in terminology occur throughout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a “weight vector” or “weights”. We employ *priority vector* here.

Specify the evidentiary span.

Figure 2 reproduces the *hierarchy model* employed by Zeng, D., et al (2025) for their pipeline-material selection. It depicts a hierarchically structured decision. The identified target describes a decisional activity: “Material selection of titanium alloy ...” Their nine criteria include eight metallurgical characteristics, $B_1, \dots, B_8$, and one economic one B_9 . The latter suggests an economic basis for utility.

The constituent criterion-to-alternative hierarchy models address how each of the candidate alloys address the metallurgical desiderata. Zeng, D., et al (2025) describe an extensive material-testing regime to ascertain the performance of each alloy with respect to the metallurgical criteria. Although comparative judgments introduce subjective aspects to the material-selection decision, Zeng, D., et al (2025) also bring to bear extensive empirical experimental evidence.

Visual, tree-graph depiction of the decision-element hierarchy aids in understanding by human decision-makers. Its idiosyncratic style notwithstanding, Figure 2 follows the ubiquitous pattern originating with Saaty (1986, 1994), and appearing throughout the MCDM literature. Structural elements of Figure 2 map to key DQ links from Figure 1.

- The rectangular box to the left “Target: material selection ...” at the left represents the *Goal or Focus*, in Saaty’s (1986) terms. A clear goal or focus results from thoughtful *Framing*, in DQ terms.
- The horizontally centered box identifies *Criteria*. Corresponding to DQ’s *Values*, these define the span of measurable criteria believed to materially distinguish between alternatives. Zeng, D., et al (2025) identified nine such criteria.
- Figure 2 explicitly depicts the *Alternatives* under consideration. That only three — distinguished by labels TA_3 , TA_{10} , and TA_{36} , each corresponding to a distinct Titanium alloy (Zeng, D., et al (2025) Table 1) — are considered makes their explicit representation here practical. When the number of alternatives is large — such as analyzing the risk of a large portfolio of distinct assets — showing all of them clutters the hierarchy-model depiction.

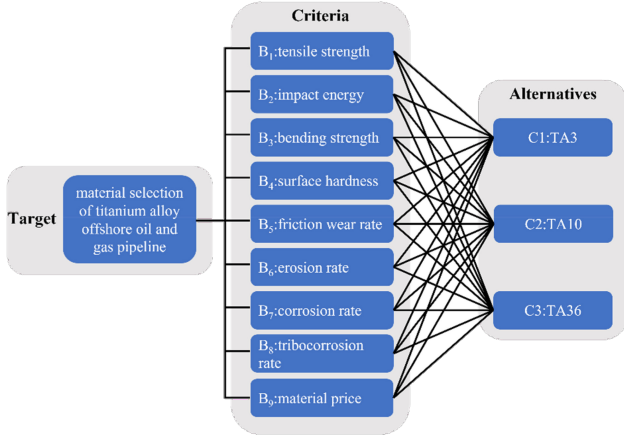
That the target-to-criterion hierarchy model in Figure 2 spans nine elements introduces a modicum of complexity. Saaty (1986) specifies a transitive-consistency criterion that, analogous to but stronger than the von Neumann-Morgenstern transitivity axiom (Kochenderfer, 2015, p. 58)

$$(A \succ B) \wedge (B \succ C) \Leftrightarrow A \succ C, \quad (1)$$

proscribes circular reasoning.

Enforcing this constraint during the comparative-judgement process can prove challenging, particularly as the number of hierarchy-model elements grows. Conventionally, AHP practitioners calculate a *Consistency Index*, which algebraically quantifies the degree to which this transitivity is satisfied (Pant, et al (2022)). Representing comparative judgments as a directed graph provides an unambiguous mechanism to identify violations of von Neumann-Morgenstern transitivity.

Figure 2: Hierarchy model used for offshore-pipeline material-selection case study. (From Zeng, D., et al (2025), Figure 9)



Alignment of AHP to mainstream decision science involves specification of a utility function. AHP goals do not by default correspond to utilities in the conventional, decision-science sense. AHP practitioners can impose such an association exogenous to the mechanics of the methodology itself. Zeng, D., et al (2025) speak to utility qualities, but do not explicitly articulate a utility function or theory.

Perform comparative judgements about elements' relative importances.

AHP imbues values and tradeoffs into decision-making through comparative judgements: "...a matrix to carry out pairwise comparisons of the relative importance of the elements..." (emphasis added) (Saaty (1986)). Pairwise is the operative term, here. Practically, one assembles a list of unique, unordered element pairs for each hierarchy model. Frequently, explicit representation of hierarchy-element comparative judgements is skipped in practice. The results are applied directly to a reciprocal matrix. Since comparative judgements themselves provide our excursion into graph-based evidentiary reasoning, we explicitly represent them here.

Comparative-judgement principles.

If the hierarchy model for a given objective (or criterion) spans n elements, then $\binom{n}{2}$ element pairs must be considered. Application of comparative judgement imposes an ordering that must subsequently be observed. Consider a hierarchy model $\mathcal{H}$ comprised of n elements, $\mathcal{H} \supseteq \{e_1, \dots, e_n\}$. Comparative judgement $\mathcal{J}_{\mathcal{H}}$ entails

$\binom{n}{2}$ distinct mappings

$$(e_i, e_j) : \mathcal{J}_{\mathcal{H}} \mapsto a_{i,j} > 0 \forall i \neq j, \quad (2a)$$

$$(e_j, e_i) : \mathcal{J}_{\mathcal{H}} \mapsto a_{j,i} = a_{i,j}^{-1}, \text{ and} \quad (2b)$$

$$(e_i, e_i) : \mathcal{J}_{\mathcal{H}} \mapsto a_{i,i} \equiv 1, \quad (2c)$$

where $a_{i,j}$ denotes the reciprocal-matrix element corresponding to (e_i, e_j) . The mapping $\mathcal{J}_{\mathcal{H}}$ imposes

$$\begin{aligned} e_i \succ e_j &\Rightarrow a_{i,j} \geq 1 \\ e_i \prec e_j &\Rightarrow a_{i,j} < 1 \end{aligned} \quad (2d)$$

Consequently, after applying the comparative judgement $\mathcal{J}_{\mathcal{H}}$, element ordering must be maintained in order to accurately interpret $a_{i,j}$. By (2a), $a_{i,j}$ is positive-definite. The $\binom{n}{2}$ distinct mappings exclude the reciprocal-matrix diagonal elements $a_{i,i}$, which by (2c) are axiomatically assigned values of unity.

Material-selection comparative judgements.

Figure 2 contains ten hierarchic models: One target-to-criterion model comprised of nine elements; and nine criterion-to-alternative models, each with three elements. Consequently, $\binom{9}{2} = 36$ element pairs must be considered for the target-to-criterion model. Each criterion-to-alternative model entails $\binom{3}{2} = 3$ element pairs. Each element pair corresponds to an edge for the graph by which the hierarchy model is represented.

Table 1a contains the comparative judgements assigned by Zeng, D., et al (2025) for each element pair depicted in the target-to-criterion hierarchy model in Figure 2. Table 1b contains element-pair judgements for each of the criterion-to-alternative models. We defer construction of corresponding reciprocal matrices to the next subsection. Explicit depiction of the comparative judgements here contributes to the clarity of present course of reasoning. In particular, the edge lists in Tables 1a and 1b provide a key of departure for subsequent graphical extension of the AHP methodology.

Assigning intensities of importance.

Now, Saaty (1986) Table 1 and Saaty (1994) Table 1 offer a "fundamental scale" containing heuristics for assignment of intensity-of-importance values to hierarchy-model element pairs. Saaty (2001b) offers a psychophysics justification for this scale, and argues for an upper bound of 9 when assigning values to importance intensities. Zeng, D., et al (2025) employ this scale (their Table 7), as do many other AHP applications appearing in the literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains

Table 1a: Comparative judgements for each element pair within the target-to-criterion hierarchy model of Figure 2. (After Zeng, D., et al (2025), Table 10.)

reference element	comparative element	intensity of importance
B_1	B_2	4
B_1	B_3	2
B_1	B_5	4
B_1	B_6	2
B_1	B_7	2
B_1	B_9	2
B_2	B_4	3
B_2	B_6	1
B_2	B_9	1
B_3	B_2	3
B_3	B_4	3
B_3	B_5	2
B_3	B_6	2
B_3	B_7	2
B_3	B_9	3
B_5	B_2	3
B_5	B_4	3
B_5	B_6	1
B_5	B_7	2
B_5	B_9	2
B_6	B_4	2
B_6	B_7	5
B_6	B_9	2
B_7	B_2	3
B_7	B_4	3
B_7	B_9	2
B_8	B_1	5
B_8	B_2	5
B_8	B_3	5
B_8	B_4	5
B_8	B_5	5
B_8	B_6	5
B_8	B_7	5
B_8	B_9	9
B_9	B_4	1

Table 1b: Comparative judgements for each element pair within each of the criterion-to-alternative hierarchy models in Figure 2. (After Zeng, D., et al (2025), Table 8).

hierarchy model	reference element	comparative element	intensity of importance
B_1	TA_3	TA_{10}	3
B_1	TA_{36}	TA_3	3
B_1	TA_{36}	TA_{10}	5
B_2	TA_{10}	TA_3	3
B_2	TA_{36}	TA_3	5
B_2	TA_{36}	TA_{10}	3
B_3	TA_3	TA_{10}	3
B_3	TA_{36}	TA_3	3
B_3	TA_{36}	TA_{10}	5
B_4	TA_3	TA_{10}	2
B_4	TA_{36}	TA_3	3
B_4	TA_{36}	TA_{10}	3
B_5	TA_3	TA_{10}	3
B_5	TA_{36}	TA_3	3
B_5	TA_{36}	TA_{10}	7
B_6	TA_3	TA_{10}	2
B_6	TA_{36}	TA_3	3
B_6	TA_{36}	TA_{10}	3
B_7	TA_3	TA_{10}	3
B_7	TA_3	TA_{36}	5
B_7	TA_{10}	TA_{36}	3
B_8	TA_3	TA_{10}	2
B_8	TA_{36}	TA_3	3
B_8	TA_{36}	TA_{10}	5
B_9	TA_3	TA_{10}	3
B_9	TA_{36}	TA_3	3
B_9	TA_{36}	TA_{10}	5

no explicit prescription for this schema for importance-intensity assignment.

Saaty (1986, 2001b) do however speak to preferential ordering, transitive consistency, and homogeneity considerations. Consequently, alternative, more-rigorous importance-intensity assignments can be considered. Application of the personal-probability indifference principle arises as one option (Savage (1972); de Finetti (1974)). Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-empirical approaches to measurement. Application of such alternatives in lieu of Saaty’s “fundamental scale” deflects a common critique of AHP’s logical rigor by recasting it terms of probabilism (Pettigrew (2016)).

Constructing comparative-judgement tables.

The construction of Tables 1a and 1b anticipates their use in defining directed graphs in the sequel. The AHP literature — Saaty (1986, 1994, 2003), in particular — prescribes nothing about the ordering of element pairs in comparative judgement. Construction of Tables 1a and 1b follows the following convention.

- Populate the “intensity of importance” column with the value greater than or equal to unity $a_{i,j} \mid (e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} \geq 1$ for each element pair (e_i, e_j) . If $(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{i,j} < 1$, then pick $(e_j, e_i) : \mathcal{J}_{\mathfrak{H}} \mapsto a_{j,i} \geq 1$ instead.
- Assign to the “reference element” column the corresponding most-preferred element $e_j \mid e_j \succ e_i$, as associated with the preferred intensity-of-importance value $a_{i,j}$. This element subsequently corresponds to the head of the directed graph edge connecting the two elements.
- Populate the “comparative element” column with the less-preferred element $e_i \mid e_i \prec e_j$. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences into the procedure. First, the structure of the table endogenously contains the directive structure of the graph. This simplifies the algorithmic logic used to construct the resultant complete graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage (1972); de Finetti (1974)) in lieu of the less-widely-recognized problematic “fundamental scale”.

Comparative-judgement indifference and its consequences.

That Table 1a contains unit-valued importance intensities — occurrences of $(e_i, e_j) : \mathcal{J}_{\mathfrak{H}} \mapsto 1$ — raises both

interest and a challenge. Saaty’s (1986) “fundamental scale” explicitly allows for this. Four undirected edges appear. This introduces an aspect of complexity. Unit-valued importance intensities imply undirected edges. In such cases, *mixed graphs* occur: Graphs containing both directed and undirected edges.

We subsequently handle this in construction of the graph Laplacians, on which entropy calculations are based. Our Laplacians must account for both directed and undirected edges, as well as other-than-unit edge weights. Aluded to previously, graph-Laplacian constructions for distinct graph classes appear separately in the literature (Wong & You (1985); Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012); Brown, et al (2020); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Judiciously combining graph-class-specific Laplacian structures with our mixed-graph structures represents the task of the present investigation.

Transform comparative judgments into priority vectors.

AHP priority vectors provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructing a reciprocal matrix from the comparative judgements for each hierarchy model. A reciprocal matrix is a Hermitian matrix with unit-valued diagonals and intensity-importance values assigned to off-diagonal elements.

Construction of reciprocal matrices from comparative judgements.

As the name implies, a reciprocal matrix is a matrix of comparative judgements within which upper-diagonal elements are reciprocals of the lower. Consider again a hierarchy model $\mathfrak{H}$ comprised of n elements, $\mathfrak{H} \supseteq \{e_1, \dots, e_n\}$. Then the corresponding reciprocal matrix $A_{\mathfrak{H}}$ becomes

$$A_{\mathfrak{H}} = \begin{bmatrix} 1 & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & 1 & \ddots & a_{2,n} \\ \vdots & \ddots & 1 & \vdots \\ a_{n,1} & a_{n,2} & \cdots & 1 \end{bmatrix} \quad (3a)$$

subject to (2b), and where values of $a_{i,j}$ are assigned according to $\mathcal{J}_{\mathfrak{H}}$ as specified by (2a and 2c).

Illustratively, the reciprocal-matrix for criterion-to-alternative hierarchy model B_1 becomes

$$A_{B_1} = \begin{pmatrix} 1 & 3 & 1/3 \\ 1/3 & 1 & 1/5 \\ 3 & 5 & 1 \end{pmatrix} \quad (3b)$$

It straightforwardly arises from applying the method above to the rows in Table 1b. Although Zeng, D., et al (2025) exhaustively represent the reciprocal-matrix set in their Table 10 and equation (13), (3a and 3b) suffice to obviate the need for complete reproduction here.

Priority-vector calculation.

The AHP-application literature contains different approaches to translating a reciprocal matrix $A_{\mathfrak{H}}$ for a hierarchy model $\mathfrak{H}$ into a priority vector $\omega_{\mathfrak{H}}$. We summarize here comparisons previously described by Saaty & Vargas (1984) and Saaty (1990).

First, that $\omega_{\mathfrak{H}}$ is a projection onto the unit simplex Δ^n where

$$\Delta^n = \left\{ x \in \mathbb{R}^n \left| \sum_i x_i = 1 \wedge x_i \geq 0 \forall i \right. \right\}$$

is highly significant. All AHP calculation results are constrained to the interval $[0, 1]$. This includes both AHP-calculation results and standardization of inputs demonstrated in a subsequent subsection.

Summary of priority-vector approaches. We summarize here comparisons previously described by Saaty & Vargas (1984). We also make passing observation of a more-empirical approach to obtaining priority vectors.

Principal-eigenvector method. Saaty (2001a, 2003) strongly argued for the principal eigenvector as the definitive approach to calculating priority vectors $\omega_{\mathfrak{H}}$. Consider the eigenvector/eigenvalue decomposition

$$A_{\mathfrak{H}} \Omega_{\mathfrak{H}} = \Omega_{\mathfrak{H}} \Lambda_{\mathfrak{H}} \quad (4a)$$

of the reciprocal matrix $A_{\mathfrak{H}}$ for a hierarchy model $\mathfrak{H}$. As usual, $\Omega_{\mathfrak{H}}$ denotes a matrix whose dimensions correspond exactly to those of $A_{\mathfrak{H}}$, and whose columns are the eigenvectors thereof. $\Lambda_{\mathfrak{H}}$ is a diagonal matrix of the corresponding eigenvalues and $\lambda_{\mathfrak{H}}$ a vector of the diagonal elements of $\Lambda_{\mathfrak{H}}$. The principal eigenvalue then appears as

$$\lambda_{\mathfrak{H}}^{\mathcal{P}} = [\lambda_{\mathfrak{H}}]_{\text{argmax}(\lambda_{\mathfrak{H}})}. \quad (4b)$$

The priority vector $\omega_{\mathfrak{H}}$ follows from the principal eigenvector as

$$[\omega_{\mathfrak{H}}^{\mathcal{P}}]_i = \frac{[\Omega_{\mathfrak{H}}]_{i, \text{argmax}(\lambda_{\mathfrak{H}})}}{\sum_i [\Omega_{\mathfrak{H}}]_{i, \text{argmax}(\lambda_{\mathfrak{H}})}}. \quad (4c)$$

Saaty (2001a) grounds the rationale for this priority-vector choice in the Perron-Frobenius Theorem (Horn & Johnson (2012); Hawkins (2021)). Priority-ordering

stability by AHP proves highly sensitive to the transitive consistency of comparative judgements. More-strenuous than von Neumann-Morgenstern transitivity (1), AHP transitivity requires

$$a_{i,j} a_{j,k} = a_{i,k}. \quad (5a)$$

Transitive consistency applies not just to preferential ordering, but also of intensities of importance.

By induction, (5a) extends to higher orders

$$a_{1,2} a_{2,3} \cdots a_{n-1,n} = a_{1,n}. \quad (5b)$$

In fact, Saaty (2001a) proves a limiting case

$$\lim_{k \rightarrow \infty} \frac{A_{\mathfrak{H}}^k \omega}{\omega^T A_{\mathfrak{H}}^k \omega} \rightarrow \omega_{\mathfrak{H}}^{\mathcal{P}} \quad (5c)$$

given some arbitrary positive-definite ω . This associates $\omega_{\mathfrak{H}}$ with the limiting case of transitive consistency specified by (5a).

Geometric-mean methods. Forman & Peniwati (1996), Csató (2019), and others argued that arithmetic or geometric means provided suitable priority vectors in some cases. Zeng, D., et al (2025) employed the geometric-mean approach

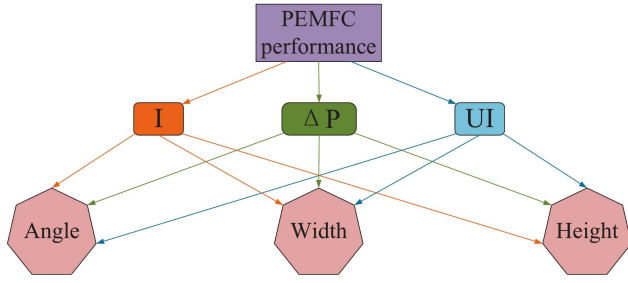
$$[\omega_{\mathfrak{H}}^{\mathcal{G}}]_i = \frac{\left(\prod_{j=1}^n a_{i,j} \right)^{\frac{1}{n}}}{\sum_{i=1}^n \left(\prod_{j=1}^n a_{i,j} \right)^{\frac{1}{n}}}. \quad (6)$$

Saaty (1990, 2001a) however demonstrated the accompanying risk of rank-reversal with this approach even under near-trivial conditions. Nonetheless, when a high degree of comparative-judgement transitivity (5a) occurs, results from (6) converge to those produced by (4c). The approach (6) offers transparency and reproducibility advantages that many researchers find convenient.

Entropy-weight method. Now, the priority-vector methods of (4c) and (6) are solely based reciprocal matrices constructed from subjective comparative judgements. Sometimes more-empirical bases for priority vectors prove useful. This allows for consideration of phenomenological information exogenous to comparative judgements.

The Entropy-Weight Method provides such an approach. Examples of its use appear in Qu, et al (2022); Shen & Wu (2023); Ren & Sun (2025). It requires a larger number of alternatives than Zeng, D., et al

Figure 3: Hierarchy model used in design of Proton-Exchange Membrane Fuel Cell (PEMFC). (From Bu, et al (2026) Figure 5)



(2025) considered. Practically, one normalizes — standardizes to a scale in $[0, 1]$ — the measurements to which priority vectors are applied. A series of calculations including entropy leads to an empirically-based priority vector.

Bu, et al (2026) applied a hybrid approach to the design of fuel cells. Figure 3 reproduces their hierarchy model. The intermediate criteria — current density I , pressure drop ΔP , Oxygen-uniformity index and UI — represent device physical-operating characteristics. The subcriteria *Angle*, *Width*, and *Height* correspond to geometric design parameters.

Bu, et al (2026) employed a physics-based computational model to establish relationships between the device-geometry subcriteria and the operating parameters. These simulation results led to an empirical basis for an entropy-based priority vector. Combining this with a priority vector based on comparative judgements produced a “hybrid” priority vector on which their design decision was based.

Now, entropy-weight priority vectors are not based constraints enforcing transitive consistency along the lines of (5a-5c). Nonetheless, exemplified by Bu, et al (2026), empirical phenomenologically-based evidence can prove more-reliable than subjective comparative judgements. The highest-quality decision depends on judicious use of the most-informative evidence available.

Priority vectors used for pipeline-material selection.

Table 2 lists the priority vectors for each hierarchy model depicted by Figure 2 based on the comparative judgements in Tables 1a and 1b. The priority vector $\omega_{\mathcal{S}}^{\mathcal{P}}$ values result from (4c). The “ $\mathbb{L}^2$ distance” column contains the Euclidean $\mathbb{L}^2$ distance between $\omega_{\mathcal{S}}^{\mathcal{P}}$ and $\omega_{\mathcal{S}}^{\mathcal{G}}$. The TA_{opt} hierarchy model corresponds to the target-to-criterion model in Figure 2.

By visual comparison between Table 2 and Figure 25 (and Table 12) from Zeng, D., et al (2025), the priority-vector values for hierarchy models $B_1, \dots, B_9$ coincide exactly. The “ $\mathbb{L}^2$ distance” column in the former indi-

cates negligible differences between $\omega_{\mathcal{S}}^{\mathcal{P}}$ and $\omega_{\mathcal{S}}^{\mathcal{G}}$. The arguments by Forman & Peniwati (1996) and Csató (2019) apparently hold up in this case.

The priority-vector values for the TA_{opt} hierarchy model differ from those reported by Table 11 from Zeng, D., et al (2025) differ from those in Table 2 in about the fifth significant digit. This is consistent with the “ $\mathbb{L}^2$ distance” column. As the hierarchy-model element span expands, achieving transitive consistency (5a) becomes increasingly difficult. Comensurately, the equivalence between (4c) and (6) becomes increasingly tenuous.

For large sets of alternatives, the divergence between priority vectors based on (4c) and (6) increases the likelihood of rank reversal for the latter. Zeng, D., et al (2025) only consider three alternatives. Provided that criterion-to-alternative priority vectors $\omega_{B_1}, \dots, \omega_{B_9}$ and input measurements to which they are applied are sufficiently distinct, the risks of rank reversal are small.

Evaluate comparative judgements for transitive consistency.

Glenn Shafer once opined, “...probability is not really about numbers; it is about the structure of reasoning...” (quoted in Pearl (1988), p. 15). The AHP methodology contains mechanisms to identify inconsistencies and contradictions in the structure of reasoning (Saaty (2003); Pant, et al (2022)). Here, we consider the types of inconsistencies in comparative judgements from which concerns might arise. We then review consistency indices, AHP’s mechanism to quantify the degree of inconsistency. Finally, we consider alternative graph-theoretic approaches to mitigate inconsistencies in reasoning.

Saaty (1986) contains an extensive set of definitions, axioms, and theorems leading to

$$\lambda_{\mathcal{S}}^{\mathcal{P}} \rightarrow n, \quad (7)$$

where n denotes the order of perfectly consistent reciprocal matrix $A_{\mathcal{S}}$ and $\lambda_{\mathcal{S}}^{\mathcal{P}}$ is defined in (4b). Being perfectly consistent, $A_{\mathcal{S}}$ invariantly satisfies (5a) and (5c).

Transitive consistency bears important decision-theoretic consequences: “...when alternatives are scaled through paired comparisons, adding a new alternative can change the relative ranking of the old ones when the judgements are inconsistent or when several criteria are used” (Saaty (1986)).

A consistency index for reciprocal matrices.

A consequence of (7),

$$CI = \frac{\lambda_{\mathcal{S}}^{\mathcal{P}} - n}{n - 1} \quad (8a)$$

Table 2: Priority vectors for pipeline-material selection case study Zeng, D., et al (2025) based on comparative judgements in Tables 1a and 1b.

hierarchy model $\mathfrak{H}$	element $e_{\mathfrak{H}_i}$	priority vector $\omega_{\mathfrak{H}}^{\mathcal{P}}$	$\mathbb{L}^2$ distance $\ \omega_{\mathfrak{H}}^{\mathcal{P}} - \omega_{\mathfrak{H}}^{\mathcal{G}}\ $
B_1	TA_3	0.2582	2.850E-32
B_1	TA_{10}	0.1047	
B_1	TA_{36}	0.6369	
B_2	TA_3	0.1047	1.560E-32
B_2	TA_{10}	0.2582	
B_2	TA_{36}	0.6369	
B_3	TA_3	0.2582	2.850E-32
B_3	TA_{10}	0.1047	
B_3	TA_{36}	0.6369	
B_4	TA_3	0.2493	2.003E-32
B_4	TA_{10}	0.1571	
B_4	TA_{36}	0.5936	
B_5	TA_3	0.2426	1.329E-32
B_5	TA_{10}	0.0879	
B_5	TA_{36}	0.6694	
B_6	TA_3	0.2493	2.003E-32
B_6	TA_{10}	0.1571	
B_6	TA_{36}	0.5936	
B_7	TA_3	0.6370	2.489E-32
B_7	TA_{10}	0.2583	
B_7	TA_{36}	0.1047	
B_8	TA_3	0.2300	7.704E-34
B_8	TA_{10}	0.1220	
B_8	TA_{36}	0.6483	
B_9	TA_3	0.2583	2.850E-32
B_9	TA_{10}	0.1047	
B_9	TA_{36}	0.6370	
TA_{opt}	B_1	0.1436	5.473E-05
TA_{opt}	B_2	0.0462	
TA_{opt}	B_3	0.1156	
TA_{opt}	B_4	0.0353	
TA_{opt}	B_5	0.0876	
TA_{opt}	B_6	0.0858	
TA_{opt}	B_7	0.0711	
TA_{opt}	B_8	0.3793	
TA_{opt}	B_9	0.0361	

provides a commonly used Consistency Index for a reciprocal matrix $A_{\mathfrak{H}}$ (Pant, et al (2022)). From (7), one obtains $CI \rightarrow 0$ for a perfectly consistent reciprocal matrix.

A Consistency Ratio

$$CR = \frac{CI}{RI} \quad (8b)$$

accompanies CI . In (8b), the denominator RI results from calculating CI for a large number of reciprocal matrices with randomly generated comparative judgements. Tabulated RI values are commonly repeated. Reproducing the common result with atypically greater numerical precision, Donhegan & Dodd (1991) reported $RI_3 = 0.4887$ for reciprocal-matrix order $n = 3$ and $RI_9 = 1.3733$ for $n = 9$. We apply these values here, leading to small disparities between reproduced calculations and those reported in Zeng, D., et al (2025).

Consistency-index values for the material-selection case study.

Table 3 lists CI (8a), CR (8b), $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ (4b), and the order of $A_{\mathfrak{H}}$ for each of the ten hierarchy models defined by Figure 2. As with the priority vectors, the values for $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ coincide almost exactly with those in Table 12 from Zeng, D., et al (2025), which actually reports $\omega_{\mathfrak{H}}^{\mathcal{G}}$. That CI and CR employ RI values based Donhegan & Dodd (1991) — which contain greater numerical precision — contributes to the small disparities. Comparable disparities occur for the target model TA_{opt} between Table 3 values and those from Zeng, D., et al (2025) Table 11.

Table 3: Consistency indices for hierarchy models defined by Figure 2 from the offshore-pipeline material-selection case study (Zeng, D., et al (2025)).

hierarchy $\mathfrak{H}$	CI (8a)	CR (8b)	$\lambda_{\mathfrak{H}}^{\mathcal{P}}$ (4b)	order $A_{\mathfrak{H}}$
B_1	0.0193	0.039	3.039	3
B_2	0.0193	0.039	3.039	3
B_3	0.0193	0.039	3.039	3
B_4	0.0268	0.055	3.054	3
B_5	0.0035	0.007	3.007	3
B_6	0.0268	0.055	3.054	3
B_7	0.0192	0.039	3.039	3
B_8	0.0019	0.004	3.004	3
B_9	0.0193	0.040	3.039	3
TA_{opt}	0.0702	0.051	9.562	9

The coincidence between $\lambda_{\mathfrak{H}}^{\mathcal{P}}$ and the order of $A_{\mathfrak{H}}$ in Table 3 provides the raw indication of the extent to which (7) is satisfied. Except for the TA_{opt} hierarchy model, the quantities coincide very closely. Pant, et al

(2022) reiterate the heuristic $CR \lesssim 0.1$ as a “good-enough” condition for most purposes. All of the CR values in Table 3 meet this condition.

Remediation of reciprocal-matrix transitive inconsistencies.

Saaty (2003) describes a procedure to “...transform a positive reciprocal matrix to a near-consistent matrix.” This involves applying perturbations to the matrix A_{ij} following a procedure described by Harker (1987). Performing such a procedure may lead to an epistemic conundrum: One applies algorithmic logic to override comparative judgements.

Saaty (2003) elsewhere argued that “When we deal with tangibles, a pairwise comparison judgment matrix may be perfectly consistent but irrelevant and far off the mark of the true values. For several reasons a modicum of inconsistency may be considered as a good thing and forced consistency without knowledge of the precise values as an undesirable compulsion.”

Order alternatives based on comparative judgements.

Entropy of edge-weighted, directed acyclic graph.

Data availability.

This GitHub repository `ahp-graph-entropy` contains the data and the python and Neo4j logic on which this paper is based.

References

- Abbas, A. & Hazen, G. (2025). “On the value of information across decision problems.” *Decision Analysis*: 22(1)1-88. <https://doi.org/10.1287/deca.2024.0187>.
- Abbas, A. (2026). “Is It Approximate? Or Is It A Fundamentally Flawed Decision-Making Method?” Society of Decision Professionals/INFORMS Decision Analysis Society webinar, May 13, 2026. May 2026 Webinar.
- Altuzarra, A.; Moreno-Jiménez, J. M.; Salvador, M. (2007). “A Bayesian prioritization procedure for AHP-group decision making.” *European Journal of Operations Research*: 182(1)367-382. <https://doi.org/10.1016/j.ejor.2006.07.025>.
- Barons, M. J.; Mascaro, S.; Hanea, A. M. (2022). “Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks.” *Risk Analysis*: 42(6):1196-1234. <https://doi.org/10.1111/risa.13772>.
- Biggs, N. (2012). *Algebraic Graph Theory*. Second Edition. Cambridge, UK: Cambridge University Press. ISBN: 9780511608704.
- Birchler, U. & Büttler, M (2007). *Information Economics*. Abington, UK: Routledge. ISBN: 9780415373456.
- Brown, D. E., et al (2020). “Entropy of tournament digraph.” *Linear Algebra and its Applications*: 584:371-393. <https://doi.org/10.1016/j.laa.2019.09.026>.
- Bu, W., et al (2026). “Multi-criteria evaluation of symmetric tesla-like valve flow channel geometry on proton exchange membrane fuel cell performance using analytic hierarchy process-entropy weight method.” *Energy Conversion and Management*: 350:120952. <https://doi.org/10.1016/j.enconman.2025.120952>.
- Camuffo, A., et al (2024a). “A scientific approach to entrepreneurial decision-making: Large-scale replication and extension.” *Strategic Management Journal*: 45(6):1209-1237. <https://doi.org/10.1002/smj.3580>.
- Camuffo, A.; Gambardella, A.; Pignataro, A. (2024b). “Theory-driven strategic-management decisions.” *Strategy Science* 9(4):297-514. <https://doi.org/10.1287/stsc.2024.0173>.
- Caporale, D. & Rinaldi, A. (2025). “The application of analytical hierarchy process to assess adaptation strategies for flood and landslides risks: A case study of a multi-risk area community.” *Environmental Science & Policy*: 163:103959. <https://doi.org/10.1016/j.envsci.2024.103959>.
- Chen, Z., et al (2015). “Entropy of weighted graph with Randić weights.” *Entropy*: 17:3710-3723. <https://doi.org/10.3390/e17063710>.
- Chen, Y., et al (2019). “Analytic network process: Academic insights and perspectives analysis.” *Journal of Cleaner Production*: 235:1276-1294. <https://doi.org/10.1016/j.jclepro.2019.07.016>.
- Cook, M. (2018). *Mathematicians: An Outer View of the Inner World*. Providence, RI: American Mathematical Society. ISBN: 9781470448387.
- Cooke, R. M. (2026). “Wisdom/Madness of Crowds and Perils of Point Forecasts.” *Decision Analysis*: article in advance. <https://doi.org/10.1287/deca.2025.0437>.

- Cover, T. M. & Thomas, J. A. (2006). *Elements of Information Theory*. Second edition. New York: Wiley. ISBN: 9780471748823.
- Cox, D. R. & Reid, N. (2000). *The Theory of the Design of Experiments*. UK: Routledge. ISBN: 9781584881957.
- Cox Jr., L. A. (2026a). "Integrating fragmented risk knowledge: Sheaf theory for risk analysts." *Risk Analysis*: 46(3):e70206. <https://doi.org/10.1111/risa.70206>.
- Cox Jr., L. A. (2026b). "Reducing risk misinformation and miscommunication: A sheaf-theoretic perspective." *Risk Analysis*: 46(4):e70224. <https://doi.org/10.1111/risa.70224>.
- Creighton, J. E. (1909). *An Introductory Logic*. London: MacMillan. LCCN: 33034164.
- Csató, L. (2019). "A characterization of the Logarithmic Least Squares Method." *European Journal of Operations Research*: 276(1):212-216. <https://doi.org/10.1016/j.ejor.2018.12.046>.
- de Finetti, B. (1974). *Theory of Probability*. Chichester, UK: Wiley. ISBN: 9781119286387.
- Dehmer, M. & Mowshowitz, A. (2011). "A history of graph entropy measures." *Information Sciences*: 181(1):57-78. <https://doi.org/10.1016/j.ins.2010.08.041>.
- Dempster, A. (1967): "Upper and lower probabilities induced by a multivalued mapping." *Annals of Mathematical Statistics*: 38(2):235-339. <https://doi.org/10.1214/aoms/1177698950>.
- Donegan, H. A. & Dodd, F. J. (1991). "A note on Saaty's random indices." *Mathematical and Computer Modelling*: 15(10)135-137. [https://doi.org/10.1016/0895-7177\(91\)90098-R](https://doi.org/10.1016/0895-7177(91)90098-R).
- Ehrig, T.; Felin, T.; Zenger, T. (2026). "Causal Reasoning and the Scientific Entrepreneur: Beyond Bayes." *Bayesian Entrepreneurship*. Cambridge, MA: The MIT Press. ISBN: 9780262052160
- Eisenmann, T. (2013, January). "Entrepreneurship: A Working Definition." *Harvard Business Review*. <https://hbr.org/2013/01/what-is-entrepreneurship>.
- Fawcett, T. (2006). "An introduction to ROC analysis." *Pattern Recognition Letters*: 27(8):861-874. <https://doi.org/10.1016/j.patrec.2005.10.010>.
- Forman, E. & Peniwati, K. (1996). "Aggregating individual judgements and priorities with the Analytical Hierarchy Process." *Proceedings of the International Symposium on the Analytic Hierarchy Process (ISAHP 1996)*, Vancouver, Canada, July 12-15, 1996, pp 383-391. <https://doi.org/10.13033/isahp.y1996.009>.
- Godsil, C. & Royle, G. (2001). *Algebraic Graph Theory*. New York: Springer. ISBN: 9781461301639.
- Guanlei, X.; Siogang, X.; Xiaotong, W. (2025). "Relative entropy based uncertainty principles for graph signals." *Signal Processing*: 227:109708. <https://doi.org/10.1016/j.sigpro.2024.109708>.
- Hagberg, A. A.; Schult D. A.; Swart, P. J. (2008). "Exploring network structure, dynamics, and function using NetworkX." *Proceedings of the 7th Python in Science Conference (Scipy2008)*. Gäel Varoquaux, Travis Vaught, and Jarrod Millman (Eds), (Pasadena, CA USA), pp. 11-15, Aug 2008. <https://doi.org/10.25080/TCWV9851>.
- Hamlett, N. A. (2019). "Competitive advantage based on information." unpublished working paper. <https://drive.google.com/file/d/1Cq7oM6BYrGWxmjf1bdhrdrORZnKBvLu4/view>.
- Hamlett, N. A. (2025). "A theory and methodology for AI-based digital transformation." *Achieving Digital Transformation through Analytics and AI*, J. Liebowitz (ed.). Singapore: World Scientific. https://doi.org/10.1142/9789811296475_0002.
- Harker, P. T. (1987). "Derivatives of the Perron root of a positive reciprocal matrix: With application to the analytic hierarchy process." *Applied Mathematics and Computation*: 22(2-3):217-232. [https://doi.org/10.1016/0096-3003\(87\)90043-9](https://doi.org/10.1016/0096-3003(87)90043-9).
- Hastie, T.; Tibshirina, R.; Friedman, J. (2009). *The Elements of Statistical Learning*. Cham, CH: Springer Nature. ISBN: 9780387848587.
- Hawkins, J. (2021). *Ergodic Dynamics*. Cham, CH: Springer Nature. ISBN: 9783030592424.
- Horn, R. A. & Johnson, C. R. (2012). *Matrix Analysis*. Cambridge, UK: Cambridge University Press. ISBN: 9780511810817.
- Howard, R. A. & Abbas, A. E (2016). *Foundations of Decision Analysis*. New York: Pearson. ISBN: P200000003532.
- Hubbard, D. W. & Seiersen, R. (2023). *How to Measure Anything in CyberSecurity Risk*. New York: Wiley. ISBN: 9781119892335.
- Hué, S. et al (2026). "Measuring the driving forces of predictive performance: Application to credit scoring." *Management Science*: online first. <https://doi.org/10.1287/mnsc.2023.02025>.

- Jackson, M. C. (2019). *Critical Systems Thinking and the Management of Complexity*. New York: Wiley. ISBN: 9781119118398.
- Jiroušek, R.; Kratochvíl, V.; & Shenoy, P. P. (2023). "Computing the decomposable entropy of belief-function graphical models." *International Journal of Approximate Reasoning*: 161:108984. <https://doi.org/10.1016/j.ijar.2023.108984>.
- Kay, J. & King, M. (2020). *Radical Uncertainty*, New York: Norton. ISBN: 9781324004776.
- Kochenderfer, M. J. (2015). *Decision Making Under Uncertainty: Theory and Application*. Cambridge, MA: MIT Press. ISBN: 9780262331708.
- Kuo, H. (2022). "Entropy of weighted graph by edge weights by using Sombor reduced index." *Journal of Corrosion and Materials*: 47:20-28. <https://doi.org/10.61336/jcm2022-3>.
- Li, H., et al (2025). "A Bayesian analysis framework for decision making with interval pairwise comparison judgements." *Decision Analysis*: 22(4)235-254. <https://doi.org/10.1287/deca.2024.0207>.
- Li, J., et al (2025). "A method for optimizing different geometric shields of D-T neutron generators by combining BP neural network and Analytic Hierarchy Process." *Computer Physics Communications*: 307:109397. <https://doi.org/10.1016/j.cpc.2024.109397>.
- Lindley, D. V.; Tversky, A.; Brown, R. V. (1979): "On the Reconciliation of Probability Assessments." *Journal of the Royal Statistical Society. Series A (General)*: 142(2)146-180. <https://doi.org/10.2307/2345078>.
- Linkov, I., et al (2006). "Multicriteria decision analysis: A comprehensive decision approach of management of contaminated sediments." *Risk Analysis*: 26(1):61-78. <https://doi.org/10.1111/j.1539-6924.2006.00713.x>.
- Liu, S., et al (2022). "Relative entropy of distance distribution based similarity measure of nodes in weighted graph data." *Entropy*: 24:1154. <https://doi.org/10.3390/e24081154>.
- Lukinskiy, Valery; Lukinskly, Vladislav; Bazhina, D. (2025). "Refining the analytic hierarchy process: A statistical and methodological exploration of expert judgments." *Alexandria Engineering Journal*: 125:526-536. <https://doi.org/10.1016/j.aej.2025.04.022>.
- Martin, R. L., & Osberg, S. (2007). Social entrepreneurship: The case for definition. *Stanford Social Innovation Review*, 2007 (Spring). <https://www.ngobiz.org/picture/File/Social%20Entrepreneur-The%20Case%20of%20Definition.pdf>.
- Merrick, J. R. W.; van Dorp, J. R.; Singh, A. (2005). "Analysis of correlated expert judgements from extended pairwise comparisons." *Decision Analysis*: 2(1):1-65. <https://doi.org/10.1287/deca.1050.0031>.
- Moon, J. W. (2015). *Topics on Tournaments in Graph Theory*. Dover. ISBN: 9780486804736. (Originally, New York: Holt, Rinehart, and Winston 1968).
- Mowshowitz, A. & Dehmer, M. (2012). "Entropy and the complexity of graph revisited." *Entropy*: 14:559-570. <https://doi.org/10.3390/e14030559>.
- Nishizawa, K. (1999). "A method to improve consistency of the comparison matrix based on iteration." *Proceedings of the International Symposium on the Analytic Hierarchy Process (ISAHP 1999)*, Kobe, Japan, August 12-14, 1999, pp 89-95. <https://doi.org/10.13033/isahp.y1999.004>.
- Olive, D. J. (2017). *Linear Regression*. Cham, CH: Springer Nature. <https://doi.org/10.1007/978-3-319-55252-1>.
- Pant, S. et al (2022). "Consistency Indices in Analytic Hierarchy Process: A Review." *MDPI Mathematics*: 10(8):1206. <https://doi.org/10.3390/math10081206>.
- Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Elsevier. ISBN: 9780080514895.
- Pettigrew, R. (2016). *Accuracy and the Laws of Credence*. Oxford, UK: Oxford University Press. ISBN: 9780198732716.
- Qu, W., et al (2022). "Entropy-Weight-Method-Based Integrated Models for Short-Term Intersection Traffic Flow Prediction." *Entropy*: 24(7):849. <https://doi.org/10.3390/e24070849>.
- Ren, Q. & Sun, M. (2025). "Using AHP-Entropy method to explore the influencing factors of spatial demand of EVs public charging stations: A case study of Jinan, China." *Journal of Cleaner Production*: 491:144779. <https://doi.org/10.1016/j.jclepro.2025.144779>.
- Rindova, V. P. & Martins, L. L. (2024). "The imagination advantage: Why and how strategists combine knowledge and imagination in developing theories." *Strategy Science*: 9(4): 297-314. <https://doi.org/10.1287/stsc.2024.0184>.

- Rosiak, D. (2022). *Sheaf Theory through Examples*. Cambridge, MA: The MIT Press. ISBN: 9780262362375. <https://doi.org/10.1287/inte.24.6.19>.
- Saaty T. L. & Vargas, L. G. (1984). "Inconsistency and rank preservation." *Journal of Mathematical Psychology*: 28(2):205-214. [https://doi.org/10.1016/0022-2496\(84\)90027-0](https://doi.org/10.1016/0022-2496(84)90027-0).
- Saaty, T. L. (1986). "Axiomatic foundation of the Analytic Hierarchy Process." *Management Science*: 32(7)841-855. <https://doi.org/10.1287/mnsc.32.7.841>.
- Saaty, T. L. (1987). "Risk — Its priority and probability: The Analytic Hierarchy Process." *Risk Analysis*: 7(2):159-172. <https://doi.org/10.1111/j.1539-6924.1987.tb00980.x>.
- Saaty, T. L. (1990). "Eigenvector and logarithmic least squares." *European Journal of Operational Research*: 48(1):156-160. [https://doi.org/10.1016/0377-2217\(90\)90073-K](https://doi.org/10.1016/0377-2217(90)90073-K).
- Saaty, T. L. (1994). "How to make a decision: The Analytic Hierarchy Process." *Applied Analytics*: (24)6:19-43. <https://doi.org/10.1287/inte.24.6.19>.
- Saaty, T. L. (2001). "Decision-making with the AHP: Why is the principal eigenvector necessary?" *Proceedings of the International Symposium on the Analytic Hierarchy Process (ISAHP 2001)*, Berne, CH, August 2-4, 2001, pp 384-396. <https://doi.org/10.13033/isahp.y2001.029>.
- Saaty, T. L. (2001). "Deriving the AHP 1-9 scale from first principles." *Proceedings of the International Symposium on the Analytic Hierarchy Process (ISAHP 2001)*, Berne, CH, August 2-4, 2001, pp 397-401. <https://doi.org/10.13033/isahp.y2001.030>.
- Saaty, T. L. (2003). "Decision-making with the AHP: Why is the principal eigenvector necessary." *European Journal of Operational Research*: 145(1):85-91. [https://doi.org/10.1016/S0377-2217\(02\)00227-8](https://doi.org/10.1016/S0377-2217(02)00227-8).
- Saaty, T. L. & Vargas, L. G. (2013). *Decision Making with the Analytic Network Process*. New York: Springer. ISBN: 9781461472797.
- Sáenz-Royo, C. & Chiclana, F. (2026). "Contribution of the ratio scale of expert judgments in the analytic hierarchy process." *Expert Systems with Applications*: 296(C):128938. <https://doi.org/10.1016/j.eswa.2025.128938>.
- Saliconi, S. & Prioli, M (2018). *Measuring Uncertainty within the Theory of Evidence*. Cham, CH: Springer Nature. ISBN: 9783319741390.
- Savage, L. (1972). *The Foundations of Statistics*. Garden City, NY: Dover. ISBN: 9780486137100. (Originally Savage, L. (1954). *The Foundations of Statistics*. New York: Wiley.)
- Shafer, G. (1976). *A Mathematical Theory of Evidence*. Princeton, NJ: Princeton University Press. ISBN: 9780691100425.
- Shen, Z. & Wu, K. (2023). "AHP-entropy Weight Method Based Comprehensive Evaluation Approach for Distribution Network Research." *Proceedings of the 3rd International Conference on New Energy and Power Engineering (ICNEPE)*, November 24-26, 2023, Huzhou, China. <https://doi.org/10.1109/ICNEPE60694.2023.10429261>.
- Shenoy, P. P. (2023). "Making inferences in incomplete Bayesian networks: A Dempster-Shafer belief-function approach." *International Journal of Approximate Reasoning*: 160(2023):108967. <https://doi.org/10.1016/j.ijar.2023.108967>.
- Society of Decision Professionals. "Decision Quality". <https://www.decisionprofessionals.com/who-we-are/decision-quality>. Accessed May 27, 2026.
- Spetzler, C. S & Staël Von Holstein, C. S. (1975). "Exceptional Paper—Probability Encoding in Decision Analysis." *Management Science*: 22(3):340-358. <https://doi.org/10.1287/mnsc.22.3.340>.
- Spetzler, C.; Winter, H.; Meyer, J. (2016). *Decision Quality Value Creation from Better Business Decisions*. New York: Wiley. ISBN: 9781119176657. <https://doi.org/10.1016/j.jlp.2019.104024>.
- Sun, G.; Kroesen, M.; Rezael, J. (2025). "Anchoring bias in value function elicitation within multiattribute value theory." *Decision Analysis*: 22(4):284-304. " <https://doi.org/10.1287/deca.2024.0308>.
- Swanson, E. B. (2024). "Technology entrepreneurship is more than one might think." *Information and Organization*: 34:100512. <https://doi.org/10.1016/j.infoandorg.2024.100512>.
- Syed, Z.; Shabarchin, O.; Lawryshyn, Y. (2020). "A novel tool for Bayesian reliability analysis using AHP as a framework for prior elicitation." *Journal of Loss Prevention in the Process Industries*: 64:104024. <https://doi.org/10.1016/j.jlp.2019.104024>.
- Townsend, D. M.; Hunt, R. A.; Rady, J. (2024). "Chance, probability, and uncertainty at the edge of human reasoning: What is Knightian uncertainty?" *Strategic Entrepreneurship Journal*: 18(3)451-474. <https://doi.org/10.1002/sej.1516>.

- Townsend, D. M., et al (2023). "Are futures computable? Knightian uncertainty and artificial intelligence." *Academy of Management Review*. (50)2:415:440. <https://doi.org/10.5465/amr.2022.0237>.
- Vaidya, O. S. & Kumar S. (2006). "Analytical hierarchy process: An overview of applications." *European Journal of Operational Research*: 169(1):1-29. <https://doi.org/10.1016/j.ejor.2004.04.028>.
- Virtanen, P., et al (2020). "SciPy 1.0: fundamental algorithms for scientific computing in Python". *Nature Methods*: 17:261-272. <https://doi.org/10.1038/s41592-019-0686-2>.
- von Hayek, Fredrick (1974). "The pretence of knowledge", Nobel Prize Lecture, December 11, 1974, <https://www.nobelprize.org/prizes/economic-sciences/1974/hayek/lecture/>.
- Wong, A. K. C. & You, M. (1985). "Entropy and distance of random graph with application to structural pattern recognition." *IEEE Transactions on Pattern Analysis and Machine Intelligence*: PAMI-7(5):599-609. <https://doi.org/10.1109/TPAMI.1985.4767707>.
- Xiao, H., et al (2025). "A simulation optimization approach for weight valuation in analytic hierarchy process." *European Journal of Operations Research*: 321(3):851-864. <https://doi.org/10.1016/j.ejor.2024.10.018>.
- Zellweger, T. & Zenger, T. (2023). "Entrepreneurs as scientists: A pragmatist's approach to producing value out of uncertainty." *Academy of Management Review*: 48(3):397-408. <https://doi.org/10.5465/amr.2020.0503>.
- Zeng, D., et al (2025). "Material selection of titanium alloy pipelines considering multi-criteria in an acidic environment based on analytic hierarchy process." *International Journal of Pressure Vessels and Piping*: 2016:105461. <https://doi.org/10.1016/j.ijpvp.2025.105461>.